The Fine Structure Constant as a Spectral Coincidence on the Hopf Fibration

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We observe that the fine structure constant α satisfies the cubic equation $\alpha^3 - K\alpha + 1 = 0$, where the coefficient $K = \pi(42 + \pi^2/6 - 1/40) = 137.036064$ is constructed from spectral invariants of the Hopf fibration $S^1 \to S^3 \to S^2$: the degeneracy-weighted Casimir trace of the S^2 base ($B = 42$), the spectral zeta function of the S^1 fiber ($F = \zeta(2) = \pi^2/6$), and a boundary correction from the S^3 total space ($\Delta = 1/40$). The selection principle $B/N = \dim(S^3) = 3$ is proven algebraically to select $n_{\max} = 3$ uniquely. The resulting $\alpha^{-1} = 137.036011$ agrees with experiment to 8.8×10^{-8} relative error with zero free parameters. A combinatorial search over 1.92×10^9 formulas built from the same spectral ingredients yields a p-value of 5.2×10^{-9}, establishing that this precision is very unlikely to be accidental. The cubic is shown to be the characteristic polynomial of a traceless $\mathbb{Z}_3$-symmetric circulant matrix, suggesting democratic coupling among the three bundle components. However, the combination rule $K = \pi(B + F - \Delta)$ is a numerical observation, not a conjecture: an empirical coincidence with structural support, not a first-principles derivation. An internal Sprint-A structural check (2026-04-18; see §, §, and the residual discussion in the Open Questions) sharpens three features: the external π prefactor is the Hopf-bundle measure $\mathrm{Vol}(S^2)/4 = \mathrm{Vol}(S^3)/\mathrm{Vol}(S^1)$; the cutoff $n_{\max} = 3$ is triply selected (the $B/N = \dim(S^3)$ principle, the unique $(+, +, -)$ sign pattern, and the agreement of two independent canonical Δ forms at the same integer); and the $(+, +, -)$ pattern is shape-compatible with Atiyah–Patodi–Singer eta-invariant boundary subtraction but is not derivable from any specific spectral-action functional [22]. The working hypothesis that emerges is that α is a *projection constant* between three structurally distinct spectral regimes, rather than a quantity derivable from a single principle.

INTRODUCTION

Richard Feynman called the fine structure constant $\alpha = e^2/(4\pi\epsilon_0\hbar c) \approx 1/137.036$ "one of the greatest damn mysteries of physics" [1]. Despite a century of quantum theory, α remains an unexplained input to the Standard Model. Attempts to derive it numerically—from π, e, prime numbers, or Platonic solids—have uniformly failed [2, 3]. Eddington's "proton–electron mass ratio" argument and Wyler's symmetric-space formula [12] both achieved suggestive precision but were criticized for lacking a physical mechanism connecting the mathematical structure to electromagnetic coupling [13].

The present work is situated within the GeoVac program, which established that the discrete graph Laplacian for hydrogen is mathematically equivalent to the Laplace–Beltrami operator on the unit three-sphere S^3, related to the Schrödinger equation via Fock's 1935 stereographic projection [4]. The electron's bound-state manifold *is* S^3—a result supported by 18 independent symbolic proofs [4].

This three-sphere carries a unique fiber bundle structure: the Hopf fibration [8, 9]

$$S^1 \longrightarrow S^3 \longrightarrow S^2. \tag{1}$$

The base S^2 parametrizes photon propagation directions, while the fiber S^1 encodes the $U(1)$ electromagnetic gauge phase. We identify a formula for α built entirely from the spectral invariants of these three manifolds, validate it statistically against a large space of alternative formulas, and identify structural underpinnings (a $\mathbb{Z}_3$-symmetric circulant matrix) that explain the cubic form. We also characterize precisely what remains unestablished.

We state clearly at the outset: *this is an empirical formula with structural support, not a first-principles derivation.* The combination rule that assembles the spectral invariants into the coupling constant K is observed, not derived.

An earlier version of this work [11] attempted to extract α from a "symplectic impedance" ratio on the electron lattice, achieving 6.2×10^{-5} relative error with a fitted helical pitch. The present result eliminates all fitted parameters and improves precision by a factor of 500.

THE HOPF BUNDLE AS ELECTRON–PHOTON GEOMETRY

Fock's 1935 stereographic projection maps the hydrogen momentum eigenvalue problem onto the unit three-sphere [10]. The eigenvalues of the Laplace–Beltrami operator on S^3 are

$$\lambda_n = -(n^2 - 1), \qquad n = 1, 2, 3, \ldots \tag{2}$$

with degeneracy $g_n = n^2$, reproducing the hydrogen spectrum upon projection to flat space via the energy-shell constraint $p_0^2 = -2E$ [4]. This identification is exact in the continuum limit: the graph Laplacian on the discrete lattice converges to the continuum operator on S^3, whose spectrum is $-(n^2-1)$, as verified by 18 independent symbolic proofs [4].

The three-sphere admits a unique fiber bundle structure, the Hopf fibration (1). Concretely, the map $\pi : S^3 \to S^2$ sends each point $(z_1, z_2) \in \mathbb{C}^2$ with $|z_1|^2 + |z_2|^2 = 1$ to the ratio $z_1/z_2 \in S^2$. Each fiber $\pi^{-1}(p)$ is a great circle $S^1 \subset S^3$.

We interpret the three components physically:

- **Total space S^3:** The electron bound-state manifold, as established by the Fock projection.
- **Base S^2:** The photon momentum sphere, parametrizing propagation directions of the electromagnetic field.
- **Fiber S^1:** The $U(1)$ gauge phase of the electromagnetic potential.

This identification is natural: the Hopf fibration is the geometric expression of $U(1)$ gauge symmetry acting on the electron's state space. Electromagnetic coupling is the "cost" of projecting S^3 onto S^2—the information lost in the fiber.

Spectral-triple setting and published lineage

The graph-spectral construction used below places naturally in the framework of *finite spectral triples on graphs* developed by Marcolli and van Suijlekom [19]. In their framework a *gauge network* is a finite spectral triple whose algebra is a direct sum of matrix algebras indexed by the vertices of a graph, with a connection variable assigned to each edge; the spectral action $\mathrm{Tr}\, f(D/\Lambda)$ on such a spectral triple reproduces Wilson lattice gauge theory on the underlying graph. The discrete S^3 Hopf graph of Paper 7, with the $U(1)$ and $SU(2)$ gauge extensions developed elsewhere in the GeoVac corpus, is a specific instance of this general construction. The Perez-Sánchez correction [21] clarifies that the continuum limit of the original gauge networks is Yang–Mills without Higgs (the companion quiver extension [20] obtains Yang–Mills–Higgs for its own enlarged construction).

The present paper does not derive α: it reports a numerical observation within the published graph-spectral-triple lineage. The three ingredients B, F, Δ each have an independent spectral home; what is GeoVac-specific is that their combination by the single rule $K = \pi(B + F - \Delta)$ lands on α^{-1}. The *machinery* is established; the *observation* is this paper's contribution. We claim no derivation of the combination rule: the open questions of §–§VII ask why the cross-sector sum closes, not whether such a framework exists.

SPECTRAL INVARIANTS OF THE BUNDLE

Each component of the Hopf bundle carries intrinsic spectral data. We extract one canonical invariant from each.

Base manifold S^2: Casimir trace

The eigenvalues of the Laplacian on S^2 are $l(l+1)$ with degeneracy $2l + 1$. These are the Casimir eigenvalues of $SO(3)$. We define the *degeneracy-weighted Casimir trace* as

$$B(n_{\max}) = \sum_{n=1}^{n_{\max}} \sum_{l=0}^{n-1} (2l + 1)\, l(l + 1), \tag{3}$$

where the outer sum runs over principal shells (each shell n contains angular momenta $l = 0, \ldots, n - 1$). This is the total angular momentum content of all electron states up to shell $n_{\max}$, weighted by the magnetic degeneracy of each subshell.

At $n_{\max} = 3$, the explicit computation gives

$$\begin{aligned} B(3) &= \underbrace{0}_{(1,0)} + \underbrace{0}_{(2,0)} + \underbrace{6}_{(2,1)} + \underbrace{0}_{(3,0)} + \underbrace{6}_{(3,1)} + \underbrace{30}_{(3,2)} \\ &= 42, \end{aligned} \tag{4}$$

where subscripts denote (n, l) pairs.

Selection principle

The total number of states through shell $n_{\max}$ is $N(n_{\max}) = \sum_{n=1}^{n_{\max}} n^2 = n_{\max}(n_{\max} + 1)(2n_{\max} + 1)/6$. We prove that the ratio

$$\frac{B(n_{\max})}{N(n_{\max})} = \dim(S^3) = 3 \tag{5}$$

is satisfied uniquely at $n_{\max} = 3$.

The inner sum in Eq. (3) has a closed form:

$$\sum_{l=0}^{n-1} (2l + 1)\, l(l + 1) = \frac{n^2(n^2 - 1)}{2}. \tag{6}$$

Therefore $B(m) = \sum_{n=1}^{m} n^2(n^2 - 1)/2$, which, using the standard power-sum identities $\sum n^2$ and $\sum n^4$, evaluates to

$$B(m) = \frac{m(m + 1)(2m + 1)(m + 2)(m - 1)}{20}. \tag{7}$$

The ratio simplifies by cancellation:

$$\frac{B(m)}{N(m)} = \frac{3(m + 2)(m - 1)}{10}. \tag{8}$$

Setting $B/N = 3$ gives $(m+2)(m-1) = 10$, i.e., $m^2+m-12 = (m+4)(m-3) = 0$, whose unique positive root is $m = 3$. This completes the algebraic proof: $n_{\max} = 3$ is the *only* positive integer at which the selection principle holds.

$n_{\max}$	$B(n_{\max})$	$N(n_{\max})$	B/N
1	0	1	0.000
2	6	5	1.200
3	42	14	3.000
4	162	30	5.400
5	462	55	8.400

TABLE I. The selection principle $B/N = 3$ is satisfied uniquely at $n_{\max} = 3$, as proven algebraically via Eq. (8).

This result connects to the Casimir structure of $SO(4)$. In the Peter–Weyl decomposition of $L^2(S^3)$, each shell n carries the (j, j) representation of $SO(4) \cong (SU(2) \times SU(2))/\mathbb{Z}_2$ with $j = (n-1)/2$. The $SO(3) \subset SO(4)$ Casimir has per-shell average

$$\langle C_3 \rangle_n = \frac{1}{n^2} \sum_{l=0}^{n-1} (2l+1)\, l(l+1) = \frac{n^2-1}{2} = C_4(n), \quad (9)$$

which equals the $SO(4)$ Casimir eigenvalue. The selection principle $B/N = 3$ is therefore equivalent to requiring that the degeneracy-weighted average of the $SO(4)$ Casimir over shells $1 \le n \le n_{\max}$ equals $\dim(S^3)$.

The base manifold invariant is

$$\boxed{B = \sum_{n=1}^{3} \sum_{l=0}^{n-1} (2l+1)\, l(l+1) = 42.} \quad (10)$$

Verification. The ingredient identities of this section are pinned by exact symbolic tests (no floats): `tests/test_paper2_ingredients.py` recomputes $B(3) = 42$ term-by-term from Eq. (3), proves the closed forms (6)–(8) symbolically, and confirms that the selection $B/N = 3$ holds at $n_{\max} = 3$ and fails at $n_{\max} \in \{1, 2, 4, 5\}$ (Table I). The generalized S^d identity of § is pinned separately in `tests/test_paper2_corrections.py`.

Triple selection at $n_{\max} = 3$

The Sprint-A internal verification [22] makes an observation that sharpens the selection principle above. The cutoff $n_{\max} = 3$ is selected not by one independent condition but by three mutually independent ones, all converging at the same integer:

1. The Casimir selection principle $B(m)/N(m) = \dim(S^3) = 3$, proven above (Eq. 8 and Table I).
2. The sign pattern of the combination rule. The eight sign assignments $\pi(\pm B \pm F \pm \Delta)$ at $m = 3$ give relative errors from α^{-1} spanning seven orders of magnitude; the $(+, +, -)$ pattern uniquely hits α^{-1} within 4.8×10^{-7} relative, with the next-best $(+, +, +)$ pattern at 1.1×10^{-3}—a separation of 3.5 orders of magnitude at $m = 3$, and no analogous sign-pattern hit at any other integer m.
3. The agreement of two canonical Δ forms. The boundary product $\Delta(m) = 1/(|\lambda_m| \cdot N(m-1))$ used below and the Dirac single-level degeneracy form $\Delta(m) = 1/g_m^{\mathrm{Dirac}} = 1/[2(m+1)(m+2)]$ identified in Phase 4H (see §) are in general distinct rational functions of m; they agree at $m = 3$ (both giving 1/40) and at no other integer cutoff.

These three mechanisms are independent: the first is representation-theoretic; the second is numerical and uses all three K-ingredients; the third tests agreement of two independently derived invariants. Their convergence at a single finite integer is stronger support for the selection of $n_{\max} = 3$ than any one alone. It does not, however, derive the combination rule $K = \pi(B + F - \Delta)$ itself, which remains a numerical observation per Theorem 1.

Fiber S^1: Spectral zeta function

The eigenvalues of the Laplacian on S^1 (a circle of circumference 2π) are k^2 for $k \in \mathbb{Z}$. The spectral zeta function is

$$\zeta_{S^1}(s) = 2 \sum_{k=1}^{\infty} k^{-2s} = 2\, \zeta_R(2s), \quad (11)$$

where ζ_R is the Riemann zeta function. At $s = 1$:

$$F = \zeta_R(2) = \frac{\pi^2}{6}. \quad (12)$$

This is the zeta-regularized spectral invariant of the photon gauge phase circle. Equivalently, F is a Dirichlet series of the Fock shell degeneracy, $F = D_{n^2}(d_{\max}) := \sum_{n \ge 1} n^2\, n^{-d_{\max}} = \zeta_R(2)$, where n^2 is the n-th S^3 shell degeneracy and $d_{\max} = 4$ is the packing lattice maximum degree (Paper 0), linking F to both the Fock projection and the packing axioms.

Total space S^3: Boundary correction

The total space S^3 contributes a finite-size correction from the interplay of its eigenvalue spectrum (2) with the cumulative state count. At the selected cutoff $n_{\max} = 3$, the highest eigenvalue is $\lambda_3 = -(3^2 - 1) = -8$ and the cumulative state count through the first non-trivial shell is $N(2) = \sum_{n=1}^{2} n^2 = 1 + 4 = 5$. The boundary term is

$$\boxed{\Delta = \frac{1}{|\lambda_3| \times N(2)} = \frac{1}{8 \times 5} = \frac{1}{40}.} \tag{13}$$

The physical significance of this boundary term is not established. It may be suggestive of UV/IR mixing on the compact manifold, but this interpretation is speculative.

Fock coupling connection. The inter-shell coupling of $\cos\chi$ in the Gegenbauer eigenbasis (Paper 7, corrected 2026-09-03) is $c^2(n,l) = \frac{1}{4}[1 - l(l+1)/(n(n+1))]$, and the inverse Fock Jacobian is $1/\Omega^4(0) = 1/16$. At the highest angular momentum of the $n_{\max} = 3$ cutoff ($n = 4$, $l = 3$) the Casimir factor is $1 - 12/20 = 2/5$, and $(2/5)/\Omega^4(0) = 1/40 = \Delta$. The boundary term thus equals a composite of two geometric quantities at the angular-momentum edge of the Paper 2 truncation (an Observation; the coupling itself is $c^2(4,3) = 1/10$).

APS-shape observation. The minus sign on Δ in $K = \pi(B + F - \Delta)$ is the only sign-flipped boundary-type correction appearing in any standard heat-kernel or index-theory decomposition on S^3 [22]. The standard Connes–Chamseddine bulk expansion $\mathrm{Tr}\, f(D/\Lambda) \sim \sum_k f_k\, \Lambda^{d-k}\, a_k$ produces an all-positive Seeley–DeWitt sequence on round S^3 (positive cutoff moments f_k, positive bulk curvature invariants a_k); the only rule in the heat-kernel / index-theory literature surveyed (Vassilevich, Connes–Chamseddine, van Suijlekom, Connes–van Suijlekom spectral truncations) that puts a $(-)$ next to bulk $(+)$ contributions is the Atiyah–Patodi–Singer eta-invariant boundary subtraction. Paper 2's $\Delta^{-1} = g_3^{\mathrm{Dirac}} = 40$ is *shape-compatible* with APS-style boundary subtraction (a finite Dirac mode count at a cutoff) but is not literally an APS eta-invariant: the η-invariant of the Camporesi–Higuchi Dirac operator on S^3 vanishes by symmetry. The APS match is therefore a shape-level structural hint, not a derivation.

Supertrace sign rule. The sign structure of $K = \pi(B + F - \Delta)$ has a cleaner reading: in the Connes–Chamseddine spectral-action framework, the *supertrace* $\mathrm{Str} = \mathrm{Tr}_S - \mathrm{Tr}_D/4$ assigns opposite signs to bosonic (scalar Laplace–Beltrami) and fermionic (Dirac) contributions via the standard $(-1)^F$ grading. B and F are both scalar-sector objects (finite Casimir trace and Fock-degeneracy Dirichlet series, respectively) and enter with $(+)$. Δ is a spinor-sector object (Dirac mode-count boundary term) and enters with $(-)$. The $(+, +, -)$ sign pattern is thus the $(-1)^F$ grading, not an ad hoc sign choice.

Euler–Maclaurin boundary reading. The identification $\Delta^{-1} = g_3^{\mathrm{Dirac}} = 40$ (Phase 4H, SM-D) gains a sharper reading from the Euler–Maclaurin (EM) decomposition of the Dirac mode-count sum. For any cutoff N, $\sum_{n=0}^{N} g_D(n) =$ integral $+ [g_D(0) + g_D(N)]/2 +$ Bernoulli corrections; the upper boundary term is $g_D(N)/2 = (N + 1)(N + 2)$. At $N = n_{\mathrm{CH}} = 3$ this equals $4 \cdot 5 = 20 = \Delta^{-1}/2$ (corrected 2026-07-03: an earlier printing equated the half-degeneracy boundary term with the full $\Delta^{-1} = 40$). The EM boundary correction thus supplies the single-level degeneracy only up to a factor of 2; the reading is suggestive — the boundary of the spectral-action sum at the selection-principle cutoff — but the identification $\Delta^{-1} = g_3^{\mathrm{Dirac}} = 40$ itself rests on the Phase 4H closed form, not on Euler–Maclaurin.

THE FORMULA AND ITS THREE HOMES

Combining the three spectral invariants of the previous section, we define the *coupling constant*:

$$K = \pi(B + F - \Delta) = \pi\left(42 + \frac{\pi^2}{6} - \frac{1}{40}\right). \tag{14}$$

Numerically,

$$K = \pi\,(42 + 1.644934 - 0.025000) = \pi \times 43.619934 = 137.036064. \tag{15}$$

Structural home of the external π. The external π prefactor in Eq. (14) has a Hopf-bundle measure identification [22]:

$$\pi \;=\; \frac{\mathrm{Vol}(S^2)}{4} \;=\; \frac{\mathrm{Vol}(S^3)}{\mathrm{Vol}(S^1)} \;=\; a_0^2, \tag{16}$$

where $a_0 = \sqrt{\pi}$ is the zeroth Seeley–DeWitt coefficient on unit S^3. The three forms are algebraically equivalent via the Hopf submersion identity $\mathrm{Vol}(S^3) = \frac{1}{2}\,\mathrm{Vol}(S^2) \cdot \mathrm{Vol}(S^1)$. The reading is "Hopf-base solid-angle per chiral charge": the integration measure of a finite-cutoff trace restricted to Hopf-base data, normalized by the Dirac chirality count. This identification categorically separates the external π (first-order Hopf-measure sector) from the internal π^2 of $F = \zeta_R(2)$ (second-order Riemann-zeta sector) and from the absence of π in B and Δ (rational sectors). Paper 18's exchange-constant taxonomy may therefore gain a new entry corresponding to the Hopf-bundle topological measure, sitting outside the existing operator-order $\times$ bundle $\times$ vertex-topology grid.

The physical α satisfies a self-consistency condition

$$\frac{1}{\alpha} + \alpha^2 = K, \tag{17}$$

which, upon multiplication by α, yields the depressed cubic

$$\boxed{\alpha^3 - K\alpha + 1 = 0.} \tag{18}$$

This cubic has three real roots (discriminant $4K^3 - 27 > 0$). The physical root is the smallest positive one, corresponding to perturbative QED ($\alpha \ll 1$).

Quantity	Value	Source
K	137.036064	Eq. (14)
α^{-1} (formula)	137.036011	Eq. (18)
α^{-1} (CODATA 2018)	137.035999...	Experiment
Relative error	8.8×10^{-8}	

TABLE II. Comparison of the Hopf bundle formula with the experimental value.

The structural question and the verdict of the Phase 4B–4I program

The three ingredients $B = 42$, $F = \pi^2/6$, and $\Delta = 1/40$ enter Eq. (14) as an additive combination. A natural question is whether they share a common spectral generator: a single construction on the geometry of the electron (here, a sector of S^3) that simultaneously produces all three. Over the period 2025–2026 we carried out an extended structural-decomposition program, documented in detail in the project's change log, to test this hypothesis across a wide range of spectral-theoretic mechanisms.

The program ran for eight phases (4B through 4I), plus the Sprint-A and Sprint K-CC follow-ons, and eliminated twelve distinct mechanisms: Hopf-twist spectral comparison on $S^1 \times S^2$; packing-π hypothesis with π forced by Paper 0 axioms; scalar Hopf graph morphism between S^3 and S^2; S^1 fiber Fock-weight traces (discrete and continuous); S^5 spectral geometry; ζ-combination search for Δ; the Standard-Model running / one-loop vacuum polarization origin for Δ; and most recently, the Dirac-on-S^3 lift of B, F, and Δ as a common generator (Tier 1, 2026).

The program produced four positive structural identifications, each of which resolves the internal content of one of the three ingredients but does not unify them:

1. $\kappa \leftrightarrow B$ via the Fock momentum-space weight (Phase 4B track α-C): the universal kinetic constant $\kappa = -1/16$ of Paper 7 has an exact algebraic relationship to the Casimir trace B at the selection cutoff $m = 3$.

2. F as the scalar Fock-degeneracy Dirichlet series at the packing exponent (Phase 4F track α-J): with $g_n = n^2$ the degeneracy of the n-th S^3 shell,
$$F \;=\; D_{n^2}(s)\Big|_{s=d_{\max}} \;=\; \sum_{n\geq 1} \frac{n^2}{n^s}\Big|_{s=4} \;=\; \zeta_R(2) \;=\; \frac{\pi^2}{6}, \tag{19}$$
where $d_{\max} = 4$ is the maximum degree in Paper 0's packing axioms.

3. Δ^{-1} as a single-level Dirac degeneracy on S^3 (Phase 4H track SM-D): with the Camporesi–Higuchi spectrum $|\lambda_n|_{\text{Dirac}} = n + 3/2$ and degeneracy $g_n^{\text{Dirac}} = 2(n+1)(n+2)$ on unit S^3,
$$\Delta^{-1} \;=\; g_3^{\text{Dirac}} \;=\; 2 \cdot 4 \cdot 5 \;=\; 40, \tag{20}$$
with the Paper-2 cutoff value $n = 3$ being the unique integer at which this identity holds.

4. $\zeta(3)$ natively in the Dirac sector at $s = d_{\max}$ (Tier 1 track D3, 2026): the Dirac analog of Eq. (19) evaluates to
$$D^{\text{Fock}}_{g_n^{\text{Dirac}}}(4) \;=\; \sum_{m\geq 1} \frac{2m(m+1)}{m^4} \;=\; 2\,\zeta_R(2) + 2\,\zeta_R(3) \;=\; \frac{\pi^2}{3} + 2\,\zeta_R(3), \tag{21}$$
a closed form mixing F with the Apéry constant $\zeta(3)$. We return to this in §.

The combined verdict of Phases 4B–4I is stated in the following summary.

Theorem 1 (Three homes, structural mystery)**.** *The three ingredients of the combination rule $K = \pi(B + F - \Delta)$ have the following canonical spectral homes on S^3:*

(B) *Finite truncated Casimir trace on the scalar Laplace–Beltrami spectrum at the selection cutoff $m = 3$, with degeneracy weight $(2l+1)\,l(l+1)$.*

(F) *Infinite Dirichlet series of the scalar Fock-shell degeneracy at the packing exponent, $F = \sum_n n^2 \cdot n^{-d_{\max}} = \zeta_R(2)$.*

(Δ) *Reciprocal of the single-level Dirac degeneracy at the selection cutoff, $\Delta^{-1} = g_3^{Dirac} = 40$.*

These three objects live in two different spectral sectors of S^3 (the scalar and spinor Laplace–Beltrami sectors); they are three different object types (finite trace, infinite Dirichlet series, single-level degeneracy); and no spectral construction on S^3 simultaneously generates all three. The K combination rule is therefore a cross-sector sum *with three separate algebraic origins, and its validity to* 8.8×10^{-8} *is a structural mystery in the sense that the sum $B + F - \Delta$ collapses to α^{-1}/π through no currently known common-generator mechanism.*

Theorem 1 is a structural statement about how each ingredient arises, not a proof of Eq. (14). The combination rule itself *remains a numerical observation.* What the Phase 4B–4I program has achieved is to move the open question from "how is each piece derived" (answered by the four positive identifications above) to "why does the cross-sector sum close to α^{-1}/π."

The three obstructions to unification

The Tier 1 Dirac-on-S^3 sprint of 2026 (tracks D1–D5) was the last of the twelve mechanisms tested. Because the Dirac sector was the most structurally promising (it already contained $\Delta^{-1} = 40$ as a clean identity), a detailed record of why it also fails to lift B and F is worth inlining. The arguments are short and closed-form.

Obstruction 1 (to lifting B). The scalar Laplacian on S^3 has a zero mode at $(n = 1, l = 0)$ whose Casimir weight $l(l+1)$ vanishes. This gives the closed-form identity

$$B(m) \;=\; \frac{m(m-1)(m+1)(m+2)(2m+1)}{20}, \qquad (22)$$

with an explicit factor $(m-1)$ enforced by the zero mode. The Dirac operator on S^3 has no zero mode: $|\lambda_n| = n + 3/2 \geq 3/2$, and the ground-level contribution $g_0^{\mathrm{Weyl}} \cdot |\lambda_0|^k$ is strictly positive for every $k \geq 0$. Cumulative Dirac/Weyl traces therefore have closed forms

$$\sum_{n=0}^{N} g_n^{\mathrm{Dirac}} \;=\; \frac{(N+1)(N+2)(2N+3)}{3}, \qquad \sum_{n=0}^{N} g_n^{\mathrm{Weyl}} \;=\; \frac{(N+1)(N+2)(2N+3)}{6}, \qquad (23)$$

neither of which carries the $(m-1)$ factor of $B(m)$. The cleanest single-point coincidence is at Paper 2's cutoff $m = 3$: $|\lambda_{m-1}| \cdot g_{m-1}^{\mathrm{Weyl}} = (7/2) \cdot 12 = 42 = B(3)$, which holds because $(m-1)(m+2) = 10$ at $m = 3$ and at no other integer m.

Paper 2's $B = 42$ therefore lives on the scalar Laplace–Beltrami sector of S^3 and does not lift to the Dirac spectrum as a universal identity. The single-point hit at $m = 3$ is a consequence of the elementary identity $(m-1)(m+2) = 10$, not a spectral theorem.

Obstruction 2 (to lifting F). The Dirac-sector Dirichlet series at $s = d_{\max} = 4$ has the closed form of Eq. (21), which mixes $F = \zeta_R(2)$ additively with $\zeta_R(3)$. The mechanism is that the Dirac degeneracy factorizes as a mixture of two homogeneity classes,

$$g_m^{\mathrm{Dirac}} \;=\; 2\,m^2 \;+\; 2\,m, \qquad (24)$$

and at $s = 4$ the m^2 subchannel produces $2\zeta(2) = 2F$ (the scalar F content) while the m subchannel produces $2\zeta(3)$ (a channel absent from the scalar degeneracy). By Apéry's theorem and standard zeta-independence results, $\zeta_R(2)$ and $\zeta_R(3)$ are $\mathbb{Q}$-linearly independent, so no rational combination of Dirac-sector Dirichlet series collapses to a rational multiple of F. The only way to isolate F from the Dirac Dirichlet series is to subtract the $2\zeta(3)$ channel by hand, which is equivalent to projecting onto the m^2 sub-weight, i.e. returning to the scalar sector.

An equivalent restatement in the Camporesi–Higuchi half-integer index: the Hurwitz-regularized Dirac spectral zeta at $s = 4$ evaluates to

$$D_{\mathrm{dirac}}^{\mathrm{CH}}(4) \;=\; \sum_{n\geq 0} \frac{2(n+1)(n+2)}{(n+3/2)^4} \;=\; \pi^2 - \frac{\pi^4}{12}, \qquad (25)$$

an inseparable mixture of π^2 and π^4 with no rational multiple of F extractable.

Paper 2's $F = \pi^2/6$ therefore lives on the scalar Fock-degeneracy arithmetic at the packing exponent and does not lift to the Dirac Dirichlet series.

Obstruction 3 (to unifying Δ with B or F). The Orduz S^1-equivariant decomposition of the Dirac spinor bundle over the Hopf fibration $S^1 \to S^3 \to S^2$ (2026, Tier 1 track D4) gives, at the Paper-2 cutoff $n_{\mathrm{CH}} = 3$, a 40-dimensional spin-Dirac level with a clean charge-parity factorization

$$\Delta^{-1} \;=\; g_3^{\mathrm{Dirac}} \;=\; 40 \;=\; \underbrace{4\cdot 5}_{\text{half-integer charges}} \;+\; \underbrace{5\cdot 4}_{\text{integer charges}} \;=\; \underbrace{20}_{\text{(+) chirali}} \qquad (26)$$

This is structurally cleaner than Paper 2's scalar factorization $\Delta^{-1} = |\lambda_3| \cdot N(2) = 8 \cdot 5$: the Dirac decomposition is a single polynomial $g_n^{\mathrm{Dirac}} = 2(n+1)(n+2)$ at a single level, whereas the scalar factorization is a product of two distinct cumulative objects. However, the Dirac refinement does not produce B or F, and the scalar "8" in Paper 2's factorization is an eigenvalue of $-\Delta_{\mathrm{LB}}$, not a Dirac/spinor quantity. The two factorizations of 40 are structurally distinct and noncombinable.

Together, obstructions 1–3 close the Dirac-on-S^3 avenue for unifying the three ingredients of K. All supporting computations are exact sympy symbolic identities; no floating-point evaluation enters the arguments above. The base spectrum/degeneracy functions live in the tested production module; the obstruction-specific identities themselves were computed in the archived sprint drivers and are recorded in the memos (they are not implemented in the production module). Sources: `geovac/dirac_s3.py`, `debug/dirac_d2_memo.md`, `debug/dirac_d3_memo.md`, `debug/dirac_d4_memo.md`, `docs/dirac_s3_verdict.md`.

Connection to the framework's rigidity theorems

Theorem 1 is consistent with, and sharpens, three established results elsewhere in the GeoVac framework:

- **Paper 7 (S^3 scalar proof chain):** the 18 symbolic proofs of Paper 7 validate the scalar Laplace–Beltrami on S^3 as the geometric home of the Coulomb spectrum. B and F both live on this sector, reinforcing that Paper 7's S^3 machinery is the ingredient-generating geometry for two of the three pieces of K.

- **Paper 23 (Fock projection rigidity theorem):** the S^3 conformal projection is unique to the $-Z/r$ potential because only $-Z/r$ produces an l-independent spectrum within each n-shell. Paper 23 deferred the Dirac-sector analog as an explicit open problem (Explorer Gap #1, Tier 1 scoping). Theorem 1 is *compatible* with that rigidity theorem but independent of it: Δ lives on the spinor spectrum as a single-level degeneracy, not as a manifestation of a Dirac Fock projection. A Dirac analog of the Fock rigidity theorem remains open and is deferred to Tier 2 or beyond.

- **Paper 24 (HO rigidity theorem):** the Bargmann–Segal lattice is bit-exactly π-free in rational arithmetic, and Paper 24 §V identifies the Coulomb/HO asymmetry as "calibration π is structurally Coulomb-specific." The Dirac-on-S^3 spectrum of Camporesi–Higuchi is *also* bit-exactly π-free in rational arithmetic (Tier 1 D1 module, 51/51 symbolic tests), and the π content in Eqs. (21) and (25) enters only through the infinite-sum analysis, not through the spectrum itself. The Dirac sector on S^3 is therefore a *structural twin* of the Bargmann–Segal HO lattice in its rational-arithmetic certification, and the Paper 2 π content in $K = \pi(B + F - \Delta)$ is the Coulomb-specific calibration of Paper 24's taxonomy.

An aside: $\zeta(3)$ in the Dirac sector

Eq. (21) is the first appearance of the Apéry constant $\zeta_R(3)$ as a structural object in the GeoVac framework. It is produced natively by the weight-m subchannel of the Dirac degeneracy $g_m^{\text{Dirac}} = 2m(m+1)$ at the Paper-0 packing exponent $d_{\max} = 4$.

We emphasize that $\zeta(3)$ is *not* part of the K combination rule. It is a *byproduct* of the Dirac sector — a transcendental that appears when one asks "what is the Dirac analog of F?" and receives the answer "F plus $\zeta(3)$, inseparably."

Structurally, $\zeta(3)$ has the following features within the framework:

- It is transcendental of unknown arithmetic type (Apéry 1978 proved irrationality; algebraic independence from π is open).

- It is not reachable from any scalar Laplace–Beltrami spectral construction on odd-dimensional spheres, as established in Phase 4E track α-I: round-sphere spectral invariants produce integer powers of π in volumes, pure rationals in Seeley–DeWitt coefficients and heat-kernel expansions, and $\zeta_R(\text{odd}) + \log 2$ in spectral determinants, but π^2 and $\zeta(3)$ never appear *additively alongside a rational.*

- It appears for the first time in the framework in Eq. (21) as the weight-m subchannel of a first-order spectral operator at the packing exponent.

This suggests — and we flag for future work — that the Paper 18 exchange-constant taxonomy (intrinsic / calibration / embedding / flow) may benefit from a distinction between *even-zeta content* (arising from second-order Laplace–Beltrami operators through Jacobi-θ inversion, which systematically lowers the power of π by 1/2 at each asymptotic order) and *odd-zeta content* (arising from first-order spectral operators through half-integer Hurwitz identities). We do not add a tier to Paper 18 here; we simply record the phenomenology.

$\zeta(3)$ has no role in reproducing α^{-1}. Its inclusion in the present section is for completeness: the Dirac-on-S^3 sector carries new content beyond what K uses, and that content is worth naming.

Closure of the structural decomposition program

With Theorem 1 and the three obstructions of §, the structural-decomposition program for α that began in 2025 and ran through Phases 4B–4I reaches its natural stopping point. The program has achieved:

1. A canonical spectral home for each of B, F, Δ, established in closed form.

2. A complete enumeration of the mechanisms by which these three ingredients might share a common spectral generator, with twelve mechanisms eliminated and no surviving candidate.

3. Identification of $\zeta(3)$ as a new framework transcendental in the Dirac sector, distinct from the Riemann-zeta-at-even-integer content of the scalar sector.

4. The Seeley–DeWitt cancellation theorem (supertrace sprint, 2026): on unit S^3, the asymptotic Dirac mode density is $g_n^{\text{Dirac}} \sim 4n^2$ while the scalar density is $g_n^{\text{scalar}} = n^2$, giving a universal ratio $a_k^{D^2}/a_k^{\Delta_{\text{LB}}} = 4 = \dim(S)$ at every Seeley–DeWitt order k, where S is the spinor bundle. The perturbative Connes–Chamseddine supertrace $\text{Str}[f(D^2/\Lambda^2)]$ therefore vanishes identically in the asymptotic expansion. All three ingredients of $K/\pi = B + F - \Delta$ are invisible to the perturbative spectral action: B is a finite Casimir trace, F is an arithmetic Dirichlet series, and Δ is an Euler–Maclaurin boundary term. K/π lives *entirely* in the non-perturbative sector of the spectral-action supertrace.

The program has *not* achieved a derivation of Eq. (14) from first principles. The agreement of

$$\pi\left(42 + \frac{\pi^2}{6} - \frac{1}{40}\right) \tag{27}$$

with α^{-1} to 8.8×10^{-8} — that is, 7 decimal digits, which no other physically-motivated formula reproduces in the extensive combinatorial search of § — remains a structural coincidence of unknown origin. We have made the mystery sharper (three spectral homes across two sectors of S^3), but we have not resolved it.

Accordingly, the combination rule retains its status as a *numerical observation*. Future work on the structural origin of α within the geometric-vacuum framework is, at the time of this writing, on hold. The next research direction in the project is the spin-full generalization of the composed qubit encoding (Paper 14), which builds on the Dirac-on-S^3 infrastructure developed in the Tier 1 sprint but is independent of the α program. That direction has a clean engineering motivation (heavy-element relativistic chemistry) and does not require the α observation to be explained.

STATISTICAL VALIDATION

A formula involving π, 42, and $\pi^2/6$ matching α^{-1} to 8.8×10^{-8} could in principle be a numerical coincidence. To test this, we performed a systematic combinatorial search [14].

Search space

The search considers formulas of the form $\alpha^{-1} = T(A+B+C)$ or, equivalently, the cubic $\alpha^3 - T(A+B+C)\alpha + 1 = 0$, where:

- A, B, C are drawn from a pool of spectral quantities: eigenvalues $|\lambda_n|$, degeneracies n^2 and $2l+1$, Casimir values $l(l+1)$, cumulative counts $N(n)$, their pairwise products up to 50, and the transcendentals $\zeta(2)$, $\zeta(3)$, $\zeta(4)$, $\ln 2$, π, π^2 (743 unique term values total).
- T ranges over 14 transcendental prefactors: π, 2π, π^2, $\pi/2$, 4π, $4\pi^2$, $1/(2\pi)$, 1, 2, 1/2, $\pi/4$, $\pi/6$, $1/\pi$, $\sqrt{\pi}$.

Both linear ($\alpha^{-1} = T \cdot S$) and cubic ($\alpha^3 - T \cdot S \cdot \alpha + 1 = 0$) interpretations are tested, giving 1.92×10^9 total formulas.

Results

At the paper's precision of 8.8×10^{-8}, only 10 formulas out of 1.92×10^9 match as well or better, yielding

$$\boxed{p\text{-value} = 5.2 \times 10^{-9}.} \tag{28}$$

The 10 formulas that beat the paper's precision involve combinations such as $4\pi(5/49 + 14/15 + \pi^2)$ that lack physical interpretation in the Hopf bundle context. The

Threshold	Hits	Fraction
10^{-2}	1,283,592	6.7×10^{-4}
10^{-3}	125,027	6.5×10^{-5}
10^{-4}	14,476	7.5×10^{-6}
10^{-5}	682	3.5×10^{-7}
10^{-6}	37	1.9×10^{-8}
10^{-7}	11	5.7×10^{-9}
8.8×10^{-8}	10	5.2×10^{-9}

TABLE III. Hit distribution across precision thresholds. The hit fraction scales approximately linearly with threshold, consistent with a uniform distribution in α^{-1} space.

paper's formula is the only one among the top matches with a coherent geometric narrative.

The p-value establishes that the precision is very unlikely to be accidental. It does not, by itself, constitute a derivation—but it does distinguish this formula from generic numerology.

THE CIRCULANT STRUCTURE

The cubic $\alpha^3 - K\alpha + 1 = 0$ is not an arbitrary polynomial ansatz. It arises naturally as the characteristic polynomial of a traceless 3×3 circulant matrix.

Circulant eigenvalue equation

Consider the traceless circulant

$$M = \begin{pmatrix} 0 & b & c \\ c & 0 & b \\ b & c & 0 \end{pmatrix}, \tag{29}$$

which has $\mathbb{Z}_3$ symmetry (invariance under cyclic permutation of rows and columns). Its characteristic polynomial is

$$\det(M - sI) = -s^3 + 3bc\, s + (b^3 + c^3) = 0. \tag{30}$$

Setting $bc = K/3$ and $b^3 + c^3 = -1$ (equivalently, $\det M = -1$), this becomes

$$s^3 - Ks + 1 = 0, \tag{31}$$

which is *exactly* the paper's cubic with $s = \alpha$. The eigenvalues of M are the three roots of the cubic. (The characteristic-polynomial shape, the tracelessness, the determinant identity $\det M = b^3 + c^3$, and the substitution to $s^3 - Ks + 1$ are verified symbolically in `tests/test_paper2_ingredients.py`, alongside the $SU(2)$ structure-constant analog below.)

Physical interpretation

The $\mathbb{Z}_3$ symmetry of the circulant represents *democratic coupling*: each of the three objects (labeled 1, 2, 3) interacts equally with the other two. In the Hopf bundle context, the three objects are the fiber S^1, the base S^2, and the total space S^3. Each component contributes a spectral invariant; the circulant structure implies that their coupling respects the cyclic symmetry of the bundle decomposition.

The $SU(2)$ structure constants ε_{ijk} exhibit the same $\mathbb{Z}_3$ cyclic symmetry, producing a circulant with $b = c = 1$ and characteristic polynomial $s^3 - 3s - 2 = 0$. The paper's cubic has the *same algebraic structure* with coupling strength $K/3 \approx 45.7$ instead of 1.

Structural content and limitations

The circulant interpretation is a *structural* result: it explains why the self-consistency equation takes the form of a depressed cubic with unit constant term. These features follow from three conditions:

1. Tracelessness: $\mathrm{tr}(M) = 0$ (eliminates the s^2 term).
2. $\mathbb{Z}_3$ symmetry: only two free parameters (b, c).
3. Unit determinant condition: $\det(M) = b^3 + c^3 = -1$ (fixes the constant term).

The condition $\det(M) = -1$ is consistent with the chirality of the Hopf map (linking number $+1$) and with the fermionic sign convention ($\psi \to -\psi$ under 2π rotation of the S^1 fiber), but neither identification constitutes a derivation.

The circulant explains *why a cubic*, but does not derive *why this specific* K. The single free parameter $bc = K/3$ encodes the coupling strength, and its value $\pi(42 + \pi^2/6 - 1/40)/3$ is not determined by the circulant structure alone.

Forced Hermiticity and the physical phase

The coupling parameters b, c are necessarily complex conjugates. Setting $s = b + c$ and $p = bc = K/3$, the constraint $b^3 + c^3 = -1$ reduces to $s^3 - Ks + 1 = 0$—the *same cubic* as for α. For real b and c, one requires $s^2 \geq 4K/3$, but all three roots satisfy $s_i^2 < 4K/3 \approx 182.7$ (the largest is $s_1^2 \approx 137.1$). Therefore [17]

$$b = \sqrt{K/3}\, e^{i\theta}, \qquad c = \sqrt{K/3}\, e^{-i\theta}, \tag{32}$$

with universal modulus $|b| = |c| = \sqrt{K/3} \approx 6.76$. The determinant condition fixes the phase:

$$\cos(3\theta) = -\frac{1}{2(K/3)^{3/2}} \approx -0.00162. \tag{33}$$

Three solutions separated by $\sim 120°$ correspond to the three roots. The physical root α has $\theta \approx 3\pi/2$ (nearly pure imaginary coupling: $b \approx -i\sqrt{K/3}$).

The circulant M is therefore *Hermitian*, not real symmetric, consistent with the $U(1)$ gauge structure of the Hopf fiber: the off-diagonal coupling carries a phase, as expected for a connection on a principal $U(1)$-bundle.

Self-referential property

The fact that $b + c$ satisfies the same cubic as α is structural, not coincidental. For any circulant of the form (29), the $k = 0$ discrete Fourier transform (DFT) mode is $\lambda_0 = b + c$, which is by construction an eigenvalue of M and therefore satisfies its characteristic polynomial. All three eigenvalues of M are the three cubic roots, regardless of which root $b + c$ equals [17].

The circulant decomposition does not factor K into independent spectral invariants of individual bundle components. The modulus $|b| = \sqrt{K/3}$ and the phase θ are both determined by K alone. The irreducible datum remains K itself.

WHAT IS AND IS NOT ESTABLISHED

Following the epistemic standards of the GeoVac program, we assess each link in the hypothetical derivation chain.

Link	Description	Status
1	Hopf bundle as geometric arena	✓
2	Spectral data $\to B, F, \Delta$	•[a]
3	Combination rule $K = \pi(B + F - \Delta)$	×
4	$\mathbb{Z}_3$ circulant $\to$ cubic	○[b]
5	α from cubic root	✓

TABLE IV. Derivation chain assessment. ✓ = established, • = largely established (see footnote), ○ = partially established, × = not established.

Link 1 (established): The Hopf fibration is the unique principal $U(1)$-bundle over S^2 with $c_1 = 1$. That S^3 is the electron state manifold is proven in Ref. [4]. Using the Hopf bundle as a geometric framework is a natural choice.

Link 2 (largely established): The selection principle $B/N = 3$ is proven algebraically to select $n_{\max} = 3$ uniquely (Sec. III). The per-shell Casimir identity $\langle C_3 \rangle_n = C_4(n) = (n^2 - 1)/2$ establishes that B is the trace of the $SO(3)$ Casimir over the truncated Peter–Weyl decomposition of $L^2(S^3)$, connecting it to the representation theory of $SO(4)$. The remaining gap is the *choice* of which spectral quantity to extract from each bundle component: why the Casimir trace for the base (not, say, the spectral determinant), why $\zeta(2)$ for the fiber (not $\zeta(1)$ or $\zeta(3)$), and why $1/(|\lambda_3| \times N(2))$ for the boundary.

Link 3 (not established): This is the critical gap. No principle selects the specific additive combination $K = \pi(B+F-\Delta)$ over alternatives such as $K = \pi B + F - \Delta$ or $K = \pi BF/\Delta$. The overall prefactor π could be argued as natural (it appears in the stereographic projection), but the additive structure is unexplained. A systematic search over spectral determinants, Ray–Singer analytic torsion, Seeley–DeWitt heat kernel coefficients, and Atiyah–Patodi–Singer invariants of the three bundle components finds no natural combination that reproduces K or K/π. The spectral determinant product $\det'(\Delta_{S^1}) \cdot \det'(\Delta_{S^3})/\pi \approx 41.957$ approaches $B = 42$ to 0.1% (see Sec. VIII) but the gap is not bridgeable by known spectral corrections. The combination rule mixes a discrete combinatorial quantity ($B = 42$) with a continuous spectral invariant ($F = \zeta(2)$) in a way that has no precedent in spectral geometry. If Link 3 can be closed, the mechanism likely involves the representation theory of $SO(4)$ rather than the analysis of smooth differential operators.

Connes–Chamseddine spectral action. The Dirac heat kernel on unit S^3 has a Seeley–DeWitt expansion that terminates at exactly two terms (by the Jacobi theta modular identity): $K(t) = (\sqrt{\pi}/2)\, t^{-3/2} - (\sqrt{\pi}/4)\, t^{-1/2} + O(e^{-\pi^2/t})$, all $a_k = 0$ for $k \geq 2$. Setting $\operatorname{Tr} e^{-D^2/\Lambda^2} = K/\pi$ gives the depressed cubic $2\Lambda^3 - \Lambda = 4(K/\pi)/\sqrt{\pi}$ with Cardano closed form $\Lambda = \sqrt[3]{A + \sqrt{A^2 - 1/216}} + \sqrt[3]{A - \sqrt{A^2 - 1/216}}$, $A = 2(B + F - \Delta)/\sqrt{\pi}$, yielding $\Lambda_\infty = 3.710245\ldots$ (50 dps). A sharp cutoff at this Λ includes exactly $40 = \Delta^{-1}$ modes, connecting the spectral action to the Euler–Maclaurin boundary reading above. However, the CC framework provides the exact functional form but *not* a mechanism to fix Λ: any target T has a unique real root (IVT + monotonicity), so the match is tautological. This confirms rather than closes the Link 3 gap.

Link 4 (partial): The cubic form arises naturally from $\mathbb{Z}_3$-symmetric coupling, and the $SU(2)$ structure constants share this symmetry. But no derivation connects the Hopf bundle to the specific circulant with $bc = K/3$.

Link 5 (established): Given the cubic, selecting the smallest positive root is pure algebra.

The structural choices that remain underived are:

1. Why additive combination $B + F - \Delta$ (not multiplicative)?
2. Why π as the overall prefactor?
3. Why $\zeta(2)$ specifically for the fiber (not $\zeta(1)$, $\zeta(3)$, etc.)?
4. Why $1/(|\lambda_3| \times N(2))$ for the boundary (not other spectral ratios)?
5. Why $\det(M) = -1$?

Three-tier composition. $K = \pi(B + F - \Delta)$ assembles three categorically different objects: B (finite Casimir, rational), F (infinite Dirichlet series at the packing exponent, transcendental), Δ (boundary product at the truncation edge, rational). Computation identifies each piece but finds no common generator. The open question is reframed from "how is each piece derived" to "why does their additive combination equal α^{-1} to 8.8×10^{-8}"—a structural conspiracy about the sum, not a derivation about the parts.

DISCUSSION

Comparison with Wyler's formula

Wyler's 1969 formula [12] also invokes symmetric spaces and achieves $\sim 10^{-5}$ precision. However, Wyler's derivation was criticized for lacking a physical mechanism connecting the symmetric space volumes to electromagnetic coupling [13]. The present work improves on this situation in two respects: (i) S^3 is the *proven* electron state manifold, providing a physical mechanism, and (ii) the p-value of 5.2×10^{-9} quantitatively distinguishes the formula from numerology. However, both approaches share the weakness of an unjustified combination rule.

Comparison with previous approach

An earlier attempt by the same author [11] modeled α^{-1} as a "symplectic impedance" ratio on the paraboloid lattice, achieving 6.2×10^{-5} relative error with a fitted helical pitch. The present result achieves 8.8×10^{-8}—a $500\times$ improvement—with zero free parameters.

Method	Error	Free params	Cutoff
Symplectic impedance [11]	6.2×10^{-5}	1	$n = 5$
Hopf bundle (this work)	$\mathbf{8.8 \times 10^{-8}}$	**0**	$n = 3$

TABLE V. Evolution of the geometric approach.

Spectral determinant near-miss

The zeta-regularized spectral determinants of the bundle components are $\det'(\Delta_{S^1}) = 4\pi^2$, $\det'(\Delta_{S^2}) = e^{1/2 - 4\zeta_R'(-1)}$, and $\det'(\Delta_{S^3}) = \pi\, e^{\zeta(3)/(2\pi^2)}$. The product

$$\frac{\det'(\Delta_{S^1}) \cdot \det'(\Delta_{S^3})}{\pi} = 4\pi^2 e^{\zeta(3)/(2\pi^2)} \approx 41.957 \quad (34)$$

is 0.1% from $B = 42$. Whether this near-miss reflects a deeper connection between the Casimir trace and spectral determinants, or is merely a coincidence, remains an open question. If a correction term derivable from the fibration geometry could bridge the 0.1% gap between $4\pi^2 e^{\zeta(3)/(2\pi^2)}$ and 42, it would provide a path from standard spectral determinants to the Casimir trace, potentially grounding Link 2 of the derivation chain (Table IV) in established spectral geometry. Replacing $B = 42$ with this determinant product makes the agreement with CODATA 11,000× *worse*, confirming that the integer Casimir trace is the correct object, not the spectral determinant.

A systematic search over additive corrections to the spectral determinant product finds the best candidates to be $+\pi/72$ (0.002% residual) and $+1/24$ (0.003% residual), but neither is derivable from first principles [16]. The quantity $\pi/72 = \pi/(|\lambda_3| \cdot n_{\max}^2)$ could be a finite-size correction, and $1/24$ appears in heat kernel coefficients, but both remain *ad hoc*. A broader search over simple arithmetic combinations of spectral determinants and fibration data finds no exact bridge. This strengthens the conclusion that the integer Casimir trace $B = 42$—not the spectral determinant—is the correct object for the base invariant.

Algebraic structure of α

The formula implies that α is algebraic over the field $\mathbb{Q}(\pi)$, since $\zeta(2) = \pi^2/6$. That is, α is a root of a polynomial whose coefficients are rational functions of π alone. If confirmed, this would constrain α far more tightly than the Standard Model currently does, and would imply that α is transcendental (since π is).

S^3 specificity

A universal algebraic identity. For any round sphere S^d, define the *formal* spectral moment

$$B_{\text{formal}}(m) \;=\; \tfrac{1}{2}\sum_{k=0}^{m} g_k\,|\lambda_k|, \qquad N(m) \;=\; \sum_{k=0}^{m} g_k, \quad (35)$$

where $\lambda_k = -k(k+d-1)$ and $g_k = \binom{k+d}{d} - \binom{k+d-2}{d}$ are the eigenvalues and degeneracies of the Laplace–Beltrami operator on S^d. At $m = 2$ these evaluate to $B_{\text{formal}}(2) = \frac{1}{2}(d+1)d(d+4)$ and $N(2) = (d+1)(d+4)/2$, giving

$$\frac{B_{\text{formal}}(2)}{N(2)} \;=\; d \qquad \text{for all } d \geq 1. \quad (36)$$

Setting $B_{\text{formal}}(m)/N(m) = d$ yields the quadratic $m^2 + dm - 2(d+2) = 0$, whose unique positive root is $m = 2$ regardless of d. The dimension-matching condition is therefore a universal property of eigenvalue moments on round spheres, not a selection principle specific to S^3.

Why the formula is specific to S^3. The quantity B appearing in the formula (Sec.) is *not* B_{formal}. It is the degeneracy-weighted trace of the $SO(3)$ *sub*-Casimir $l(l+1)$ over the $SO(4) \to SO(3)$ branching of each S^3 shell:

$$B \;=\; \sum_{n=1}^{m}\sum_{l=0}^{n-1}(2l+1)\,l(l+1). \quad (37)$$

For S^3, $B = B_{\text{formal}}$ because the diagonal $SO(3) \subset SO(4)$ embedding has index 1, giving the identity $\langle C_{SO(3)}\rangle_n = n(n+2)/2 = |\lambda_n|/2$. For all other spheres S^d with $d \neq 3$, $B \neq B_{\text{formal}}$ and $B/N = d$ has no solution at any integer m.

Three independent properties of S^3 underlie this coincidence:

1. $SO(4) \cong (SU(2) \times SU(2))/\mathbb{Z}_2$ is a product group, yielding the clean Clebsch–Gordan branching $l = 0, \ldots, n-1$ within each shell. No analogous product structure exists for $SO(d{+}1)$ with $d \neq 3$.

2. $S^3 \cong SU(2)$ is the only sphere that is a Lie group, enabling the Peter–Weyl decomposition that makes B computable as a representation-theoretic trace.

3. The embedding index of diagonal $SO(3) \subset SO(4)$ is 1. For the quaternionic Hopf fibration, the corresponding $SO(5) \subset SO(8)$ embedding through $Sp(2) \times Sp(1)$ has index ratio 5/14, which would give $B/N = 5 \neq 7$ if used as the branching factor.

The formula is not one instance of a family parametrized by d; it exists only for S^3.

Quaternionic and octonionic Hopf fibrations. Applying the combination rule $K = \pi(B + F - \Delta)$ formally to all four Hopf fibrations ($S^0 \to S^1 \to S^1$, $S^1 \to S^3 \to S^2$, $S^3 \to S^7 \to S^4$, $S^7 \to S^{15} \to S^8$) yields $1/\alpha \approx 15.4$, 137.036, 970.0, and 7164.1 respectively [18]. Only the complex Hopf value matches a known coupling constant. The quaternionic value $K/\pi \approx 309$ satisfies $K_{\text{quat}}/K_{\text{em}} \approx 7.08 \approx \dim(S^7)$, but the 1.1% deviation from 7 and the absence of any coupling at $\alpha_W \approx 1/29.5$ rule out a weak-interaction interpretation. The representation-theoretic obstructions above explain why: without the $SO(4)$ product structure, the branching-based B that makes the formula work has no natural generalization.

Dynamical specificity: the Fock projection rigidity theorem. The representation-theoretic arguments above show that the *algebraic structure* of $B = 42$ is unique to S^3. An independent *dynamical* argument reaches the same conclusion from the Fock projection itself, rather than from the $SO(4)$ branching combinatorics. The rigidity theorem (established in the nuclear extension of the GeoVac framework) states that the stereographic projection $p_0 = \sqrt{-2E_n}$ maps a one-electron central-field

Hamiltonian onto the free Laplacian on S^3 if and only if the spectrum E_{nl} is independent of l within each shell n. The Coulomb potential $-Z/r$ is the unique central potential exhibiting this accidental degeneracy, by the $SO(4)$ symmetry generated by the Runge–Lenz vector. Any deformation $V(r) = -Z/r + \delta V(r)$ with $\delta V \not\equiv 0$ lifts the l-degeneracy and breaks the conformal equivalence between the radial Schrödinger problem and the S^3 graph Laplacian. The harmonic oscillator, Woods–Saxon, square well, and Yukawa potentials all share the angular momentum algebra ($SO(3)$ Gaunt selection rules) of the Coulomb problem, but none of them carries the Fock projection because none has the $SO(4)$ accidental degeneracy. The angular sparsity of the GeoVac framework [5] is universal across spherical fermion systems; the S^3 conformal projection and the Hopf bundle structure built upon it are Coulomb-specific.

The implication for the present observation is that any derivation of α from spectral geometry must use the Coulomb-specific features of S^3 — the accidental degeneracy, the Runge–Lenz vector, and the conformal weight assignments that the Fock projection imposes — not merely the $SO(3)$ angular momentum structure that S^3 shares with other manifolds. As a numerical check on this narrowing, the analog of $B = 42$ in the nuclear shell model (j-coupled $(2j+1)\,j(j+1)$ trace at the same truncation $n_{\max} = 3$) evaluates to 103.5 and shows no coincidence with any quantity in $K = \pi(B + F - \Delta)$ at the $p < 10^{-3}$ level [7]. This is consistent with the structural argument: nuclear shell model angular momentum lives on $SO(3)$ with broken $SO(4)$, lacks the Fock projection, and therefore cannot be expected to produce the same combinatorial constants as the Coulomb-specific calculation.

The Bargmann–Segal lattice for the harmonic oscillator (Paper 24, [6]) sharpens this further: the harmonic oscillator's natural discretization on the Hardy space of S^5 is demonstrably π-free in exact rational arithmetic, with a linear projection $E_N = \hbar\omega(N + 3/2)$ that requires no calibration exchange constant. The harmonic oscillator's complex / first-order geometry produces no spectral zeta corrections at all, so no π-class transcendental enters its graph-to-physics map. The appearance of π in $K = \pi(B + F - \Delta)$ is therefore not a generic feature of quantum discretization but a specific consequence of the Coulomb potential's S^3 Riemannian (second-order) character; the harmonic oscillator's S^5 complex (first-order) character has no analog. Any future derivation of the combination rule must rely on the Riemannian/$SO(4)$ structure that distinguishes Coulomb from every other central potential.

Transition operators and the Hopf geometry

The discrete graph Laplacian for hydrogen has two types of lattice transitions [16]: $T_\pm$ (changing n by ± 1, preserving l and m) and $L_\pm$ (changing m by ± 1, preserving n and l). We observe that these correspond naturally to the Hopf fibration:

- $L_\pm$ is motion along the Hopf fiber S^1. The magnetic quantum number m is the $U(1)$ charge (eigenvalue of J_z); incrementing m rotates around the fiber circle.
- $T_\pm$ is radial motion on S^3, perpendicular to the fiber. Changing n moves between eigenspaces of Δ_{S^3}; in the Fock stereographic picture, this changes the "latitude" on S^3.

The angular momentum l is conserved by both transitions—it labels the $SO(3)$ representation within each $SO(4)$ shell but does not contribute an independent transition type. The base S^2 constrains state labeling (which l, m values exist) but does not generate lattice transitions. The Fock stereographic projection collapses the 2-dimensional base directions into a single radial variable (the principal quantum number n), explaining why $d_{\max} = 2 \times 2 = 4$ (two motion types $\times$ two directions) rather than 6.

Second selection principle

The topological maximum degree $d_{\max} = 4$ gives an asymptotic bipartite spectral radius bound of $2d_{\max} = 8$ for the graph Laplacian [15]. The S^3 Laplace–Beltrami eigenvalue $|\lambda_n| = n^2 - 1$ equals this bound uniquely at $n = 3$:

$$n^2 - 1 = 2\,d_{\max} \quad \Longrightarrow \quad n_{\max} = 3. \tag{38}$$

This provides an independent selection of $n_{\max} = 3$, from the discrete–continuous compatibility condition rather than from the Casimir selection $B/N = \dim(S^3)$.

We note a subtlety: $d_{\max} = 4$ is the topological maximum degree for interior nodes of the lattice class (first realized at $n_{\max} = 4$), not the realized maximum at $n_{\max} = 3$ (which is 3, since all 14 nodes are boundary nodes) [16]. The selection principle should be read as: the S^3 eigenvalue at $n_{\max}$ matches the *asymptotic* spectral radius bound of the lattice topology.

The connection to κ is immediate: if $\kappa = -E_1/|\lambda_{n_{\max}}| = -1/(2(n^2-1))$, then the Hopf selection $n_{\max} = 3$ gives $\kappa = -1/16$, the kinetic scale factor of the GeoVac program [4].

Status: this is a structural observation, not a derivation. The two selection principles ($B/N = 3$ and $2d_{\max} = n^2 - 1$) are algebraically independent; whether they share a common origin in the Hopf fibration geometry is an open question.

Open questions

1. *Derive the combination rule* from a path integral or spectral determinant identity on the Hopf bundle. This is the critical gap (Link 3 in Table IV). The Connes–Chamseddine spectral action on continuum S^3 provides an exact depressed cubic with Cardano closed form but does not select the cutoff Λ autonomously (see Sec. VII).

2. *Connect the $\mathbb{Z}_3$ circulant* to the differential geometry of the Hopf connection. The $SU(2)$ structure constants share the cyclic symmetry; can the coupling strength $K/3$ be derived from the connection curvature?

3. *Understand the residual* 8.8×10^{-8}. Higher-order spectral invariants (e.g., the η-invariant of S^3, which vanishes by symmetry), gravitational corrections, or weak-mixing contributions could account for the discrepancy. One-loop QED running of the formula's α from the electron mass to 1 TeV shows that the formula's value corresponds to the Thomson limit ($q^2 \to 0$), matching CODATA most closely at $\mu \approx m_e$. The residual is intrinsic to the formula, not explained by renormalization group running.

   An internal structural check [22] finds that the post-cubic residual $R_{\text{predict}} = K - \alpha^{-1} - \alpha^2$ (where the α^2 term is the trivial offset from the cubic) matches $\pi^3\alpha^3$ to within 0.25% at 80-digit precision: $R_{\text{predict}} = 1.2079 \times 10^{-5}$, $\pi^3\alpha^3 = 1.2049 \times 10^{-5}$, $R_{\text{predict}}/(\pi^3\alpha^3) = 1.0025$. The π^3 prefactor is uniquely picked out among $\{\pi^2, \pi^3, \pi^4\}$ (the other candidates are off by factors of ~ 3). A natural structural reading is $\pi^3 = \mathrm{Vol}(S^5)$, connecting to the Bargmann–Segal S^5 lattice of Paper 24 [6]. An internal cross-check computes the Seeley–DeWitt coefficients a_0, a_2, a_4 on round S^5 (each factorizes as $\pi^3 \times$ rational) and finds that no such coefficient produces a contribution at order α^3 structurally against the leading α^{-1} of K: the Connes–Chamseddine asymptotic expansion on $d = 5$ is in odd powers of Λ only, with no log term, and standard spectral-action truncations fix α once via a_0/a_4 ratios, not at order α^3. The $\pi^3\alpha^3$ match is therefore a numerical pattern of unexplained structural origin at this precision—a suggestive hint, not a derivation. Paper 24's π-free certificate on the discrete Bargmann–Segal graph is reconciled: it applies to the discrete spectrum and edge weights, not to continuum integration measures such as $\mathrm{Vol}(S^5)$.

4. *The other cubic roots.* The two non-physical roots of Eq. (18) satisfy $\alpha \approx \pm 11.7$, corresponding to $1/\alpha \approx \pm 0.085$. A systematic comparison of all natural transformations of these roots (including r, $1/r$, r^2, $\sin^2(\arctan r)$, etc.) against known Standard Model parameters ($\sin^2\theta_W$, α_s, Cabibbo angle, Yukawa couplings, mass ratios) finds no match at the 1% level. The other roots do not appear to encode additional coupling constants.

5. *Common origin of the two selection principles.* The second selection principle ($2d_{\max} = n^2 - 1$ at $n = 3$) and the Casimir selection ($B/N = 3$ at $n = 3$) are algebraically independent. Whether they share a common origin in the Hopf fibration geometry is open.

6. *Circulant phase and the Hopf connection.* The Hermiticity of the circulant coupling and its $\theta \approx 3\pi/2$ phase at the physical root may encode the $U(1)$ gauge structure. A derivation connecting the circulant phase to the Hopf connection 1-form would ground Link 4 of Table IV.

CONCLUSION

The fine structure constant satisfies the cubic equation $\alpha^3 - K\alpha + 1 = 0$, where $K = \pi(42 + \pi^2/6 - 1/40)$ is constructed from three spectral invariants of the Hopf fibration. The selection principle $B/N = \dim(S^3)$ is proven algebraically to select $n_{\max} = 3$ uniquely, via the closed-form ratio $B/N = 3(m+2)(m-1)/10$; this result is specific to S^3: a formal spectral moment $B_{\text{formal}}/N = d$ holds universally at $m = 2$ for all round spheres (Sec.), but the branching-based B that enters the formula coincides with B_{formal} only for $d = 3$, due to the unit embedding index of $SO(3) \subset SO(4)$. The quaternionic and octonionic Hopf fibrations yield $1/\alpha \approx 970$ and 7164 respectively, matching no known coupling constants. The formula contains no fitted parameters and achieves 8.8×10^{-8} relative agreement with experiment. A combinatorial search over 1.92×10^9 formulas yields a p-value of 5.2×10^{-9}, establishing that this precision is very unlikely to be accidental. The cubic form is the characteristic polynomial of a traceless $\mathbb{Z}_3$-symmetric circulant matrix, providing structural justification for the self-consistency equation.

The circulant coupling is necessarily Hermitian (b and c are complex conjugates), with the physical root at phase $\theta \approx 3\pi/2$, consistent with $U(1)$ gauge structure. A second selection principle—the bipartite spectral radius bound $2d_{\max} = 8$ matching $|\lambda_3| = n^2 - 1 = 8$—independently selects $n_{\max} = 3$ and connects to the kinetic scale factor $\kappa = -1/16$.

The combination rule $K = \pi(B + F - \Delta)$ remains a numerical observation. A systematic investigation of spectral determinants, analytic torsion, heat kernels, and index-theoretic invariants finds no smooth spectral-geometric derivation; the rule mixes discrete representation-theoretic data ($B = 42$) with continuous spectral invariants ($F = \zeta(2)$) in a way that has no known precedent. The formula has the character of a remarkable empirical observation with structural support—the spectral ingredients are natural, the cubic form is natural, the selection principle is proven, but their assembly into K awaits a first-principles derivation.

This work supersedes an earlier "symplectic impedance" approach [11] by the same author. The evolution from fitted to parameter-free is documented for intellectual transparency. The four-phase computational audit (arithmetic verification, combinatorial p-value analysis, cubic structure investigation, and derivation chain assessment) that validated the formula and identified the circulant structure is available as supplementary material [14]. A systematic investigation of spectral determinants, analytic torsion, heat kernel expansions, and index-theoretic invariants of the Hopf bundle components, documented in the project repository, established that the combination rule (Link 3) lies outside the scope of standard smooth spectral geometry. A subsequent three-phase investigation of the kinetic scale factor $\kappa = -1/16$ established the transition operator–Hopf correspondence, a second selection principle from the bipartite spectral radius bound, and the forced Hermiticity of the circulant coupling [15–17]. A further investigation applied the formula's recipe to all four Hopf fibrations, discovered the universal algebraic identity $B_{\text{formal}}/N = d$, and confirmed the S^3 specificity of the branching-based selection principle [18]. Computational implementation and documentation drafting were performed with the assistance of large language models (Anthropic Claude) under the author's scientific direction. All results were validated against analytical benchmarks.

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